Samuel Bradley Dowling

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PROFILE

Software Design with VR and Gaming graduate (BSc, Level 7, Distinction) from TUS, now completing a BSc (Honours) in the same programme. Hands-on experience in programming, web development, Unity game development, embedded systems and networking, including IoT firmware and ESP32 hardware developed on industry placement. Skilled in Java, Python, C# and C++, with practical experience using GitHub, Jira, Eclipse, VS Code, Unity, Blender and PlatformIO. Passionate about software engineering and immersive technologies, and motivated to bring strong problem-solving and teamwork skills to an innovative engineering team.

EDUCATION

BSc (Honours, Level 8) in Software Design with Virtual Reality and Gaming 2026 – Expected 2027
Technological University of the Shannon (TUS Midlands)

BSc (Level 7) in Software Design with Virtual Reality and Gaming 2023 – 2026
Technological University of the Shannon (TUS Midlands) · Final Award: Distinction (70%)

PLC (Level 5) in Computer Science 2021 – 2022
Portlaoise Institute · Final Award: Merit

PROJECTS

- **AegisMR – Final Year Project** (Unity / C# / OpenXR / Python / Scapy / Meta Quest 3S, in progress) – Developing a mixed-reality cybersecurity and networking education platform for Meta Quest 3S, using passthrough to display interactive 3D network topologies. Building a Python backend that uses Scapy to generate simulated packet traffic, streamed to Unity, with planned learning modules on network faults, Wi-Fi deauthentication and SQL injection.
- **IoT Home Security & Intrusion Detection System** (C++ / ESP32 / ESP32-CAM / PlatformIO / Flask / Cloudflare) – Designed and implemented an IoT security system for university placement, developing embedded C++ firmware for an ESP32 alarm controller and an ESP32-CAM module. Wired alarm components via GPIO, integrated livestream and snapshot capture, and built reliable end-to-end communication with a Flask backend exposed through Cloudflare to a live frontend dashboard.
- **Multiplayer Battleship Game** (C++ / SFML / GitHub) – Contributed to a team-built, turn-based multiplayer Battleship game in C++17 with TCP/IP client-server networking. Designed the UML class diagram defining the game's OOP architecture (ship inheritance hierarchy, Singleton GameManager) and authored the project brief that set its scope and requirements.
- **Web-Based Math Game** (C++ / Crow Framework / GitHub) – Developed an interactive math game using the Crow micro web framework in C++, implementing routing, UI logic and deployment workflows. Published to Oracle Academy as part of a Software Engineering module.
- **3D Unity Platformer/FPS** (C# / Unity Engine) – Built a 3D game prototype, applying Unity's physics, animation and scripting systems while strengthening C# programming and game engine architecture knowledge.
- **Agile Team Project** (Java / JUnit / Bitbucket) – Collaborated in a team of four using Agile methodologies to design, implement and test a Java application. Managed branching, merging, pull requests and CI workflows through Bitbucket.

WORK EXPERIENCE

Food and Beverage Assistant | Kelly's Steakhouse, Portlaoise July 2021 – Present

- Delivered excellent customer service for over 5 years in a fast-paced environment, strengthening communication and interpersonal skills.
- Collaborated with kitchen and front-of-house teams to streamline service and resolve issues quickly.
- Attended sustainability workshops and applied Green Skills in the workplace, supporting the company's sustainability goals.

Work Experience for PLC | Keith McLearn, Málaga, Spain Summer 2022

- Contributed to the development and maintenance of a company website using HTML, CSS, JS and WordPress.
- Communicated with colleagues, Spanish locals and the customer, and collaborated with the team from home and on site in daily stand-up meetings.

Stock and Inventory Assistant | Paragon (formerly Spicers Ltd), Citywest, Dublin 2019 – 2020

- Used the Perpetual Inventory (PI) system to monitor stock levels in real time, ensuring accuracy and reducing discrepancies.
- Applied Excel to track and update stock movements within the warehouse system.

Teen Leader (Volunteer) | MYFRC, Mountmellick, Laois 2018 – 2019

- Assisted in planning and leading group activities, developing organisational and leadership skills.
- Represented MYFRC at a Tulsa conference by addressing a live audience on camera.

ACHIEVEMENTS

- Graduated with a Distinction in a BSc (Level 7) in Software Design with VR and Gaming from TUS.
- Engineered embedded firmware and hardware for a multi-device IoT Home Security & Intrusion Detection System on TUS placement.
- Attended an Executive Leadership Course with Common Purpose and Google in Dublin, and was awarded a certificate.
- Developed a functioning website for El Taller: Yoga y Movimiento in Málaga, Spain.
- Achieved a Merit in Computer Science at Portlaoise Institute.